Writing sample

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The following writing sample is an internal proposal for a fictional SaaS-based payments platform, including cover letter, executive summary, industry overview, and technical section. I also formatted and designed the document, and I created the supporting graphs in Tableau. This proposal was a writing assignment from the Technical Communication II course through UC Berkeley Extension.

[Logo Placeholder]

MerchBill
1234 Any Street #100
Any Town, CA 91234

December 15, 2019

John Smith
Vice President, Product Development
MerchBill
1234 Any Street #100
Any Town, CA 91234

Dear Mr. Smith:

Thank you for considering our proposal for subscription management and billing (SMB) platform enhancements. We have enclosed a document detailing improvements—a push notification microservice and dunning management—to our SMB platform and their potential benefit to MerchBill and the merchants we serve.

MerchBill must move with the industry and offer our SMB services to new entrants of the subscription economy and to those merchants that are growing; however, we need a product that better fulfills merchant needs, and we have the challenge of overcoming our competitors. In terms of annual revenue, MerchBill is midtier in comparison to our closest competitors, primarily other small subscription billing and management companies such as Zuora, Aria Systems, and Recurly.

The development team and I are confident that the platform enhancements will help MerchBill gain a stronger position among our peers. We appreciate your consideration of this proposal and look forward to collaborating on this project. Should you have any further questions or concerns, I can be contacted by email at prscllm@merchbill.com or by phone at 415-555-1234.

Sincerely,

Priscilla Miller

Priscilla Miller
Technical Writer
MerchBill

Enc: (1)

Proposal: SMB Platform Enhancements

December 15, 2019

Prepared for: John Smith, Vice President, Product Development, MerchBill
Prepared by: Priscilla Miller, Technical Writer, MerchBill

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Executive Summary

Background

In recent years, the market for subscription products and services has become popular with many consumers, leading to more merchants adopting a subscription-based billing model. In turn, these merchants look to online payments platforms to support the processing of payments from their customers. However, traditional online billing platforms have limitations that can hinder a merchant's growth.

Our own subscription management and billing (SMB) platform also is impaired by these limitations; our current technology lacks two important features of a subscription billing platform: push notifications to merchants and dunning management.

A **push notification microservice** would allow merchants to receive real-time notification of subscription activity. For each event in a subscription lifecycle, such as invoice generation, attempted payments on amounts owed, and service and product calculations, the merchant would receive an alert and be able to respond to the event appropriately.

An **integrated dunning management** feature would allow for automatic emails to be sent to customers by merchants whenever payment is declined. Dunning management can help a merchant maintain a business-customer relationship and reduce churn rates.

Summary

With a solid footing in the payments industry since 2003 and experience helping organizations craft subscription billing and management solutions, MerchBill has the potential to reach more subscription merchants and help new entrants in the subscription market transition to a subscription-based business model. We currently have on staff personnel knowledgeable in the technical aspects required to implement the proposed features. Implementation should take no more than 10 months and incur costs of just under $1.4 million.

Both MerchBill and merchants will benefit from the addition of these features. Merchants will be less burdened by the challenges that come with the management of subscribers and recurring billing. MerchBill can experience decreased internal costs and will have a product that can drive us toward a stronger market position among our closest competitors.

About MerchBill

By updating our payments technology and enhancing our service capabilities, a refreshed SMB platform can help us grow to compete with the leading providers within the payments industry. With an annual revenue of about $18.3 million, MerchBill is a relatively smaller player in the overall payments processing industry, leaving us far behind the industry leader PayPal, which has an annual revenue of about $15.8 billion. MerchBill primarily competes with other smaller recurring billing companies, such as Zuora, Aria Systems, and Recurly.

Industry Overview: Financial Services

Industry Profile

Industry Description

The finance and insurance sector includes establishments primarily engaged in financial transactions (that is, transactions involving the creation, liquidation, or change in ownership of financial assets) and/or in facilitating financial transactions [10].

Industry Performance: Current and Historical

The financial services industry plays an important role within the economy by facilitating financial transactions and providing financing and risk-management services to consumers, businesses, and government agencies. As a core tenet of the economy, demand for financial services is primarily determined by prevailing economic conditions. As the economy grows, so too does demand for a variety of the sector's services. Overall, the sector has increased an annualized 2.8% to $4.7 trillion over the five years to 2018, including growth of 2.3% in 2018 alone [9]. The financial services industry revenue growth from 2008 to 2018 is shown in Figure 1.

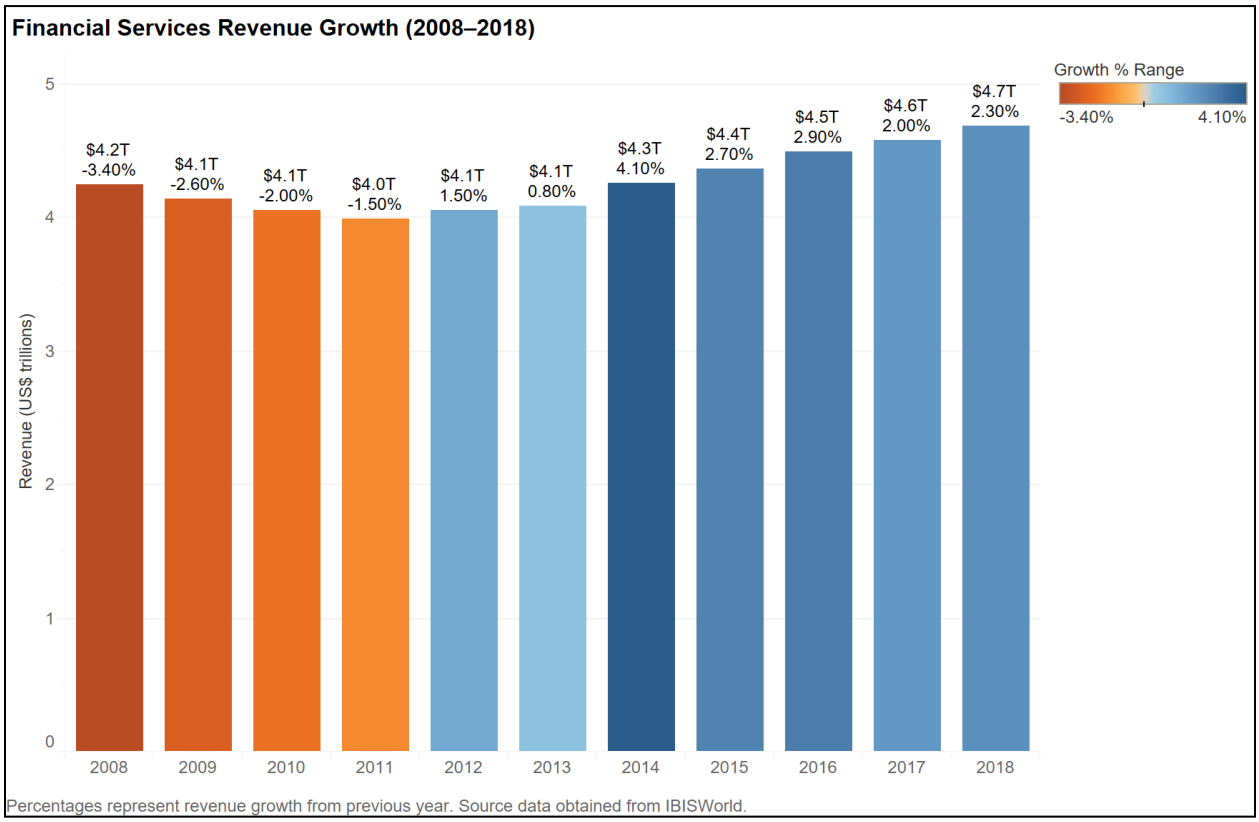

Figure 1. Financial Services Industry Revenue Growth from 2008 to 2018 [9]

In addition to the industry's stable growth, participation in the industry has also gradually increased. New operators in the sector have primarily been financial technology companies, commonly known as fintech companies [9]. The fintech companies include technology-enabled firms offering financial services and entities providing technology services directly to financial institutions and employing technology to support financial transactions among businesses and consumers. Such companies offer services within areas such as digital lending, payments processing, blockchain, e-commerce, and digital wealth management [1]. While fintech firms currently represent a small share of overall revenue of the financial services industry (approximately 6%), their growth and technological innovation is evident [2]. As a result of the rise in fintech, the number of industry enterprises has grown gradually during the period at an annualized rate of 0.6% over the five years to 2018 to 972,050 enterprises [9].

Industry Outlook and Forecast

The financial services industry is expected to continue to grow over the five years to 2023; however, growth is expected to be slightly slower as a result of normalizing demand for services. Overall, the sector is expected to increase an annualized 1.6% to $5.1 trillion over the five years to 2023 [9]. See Figure 2 for the industry's predicted revenues from 2019 to 2024.

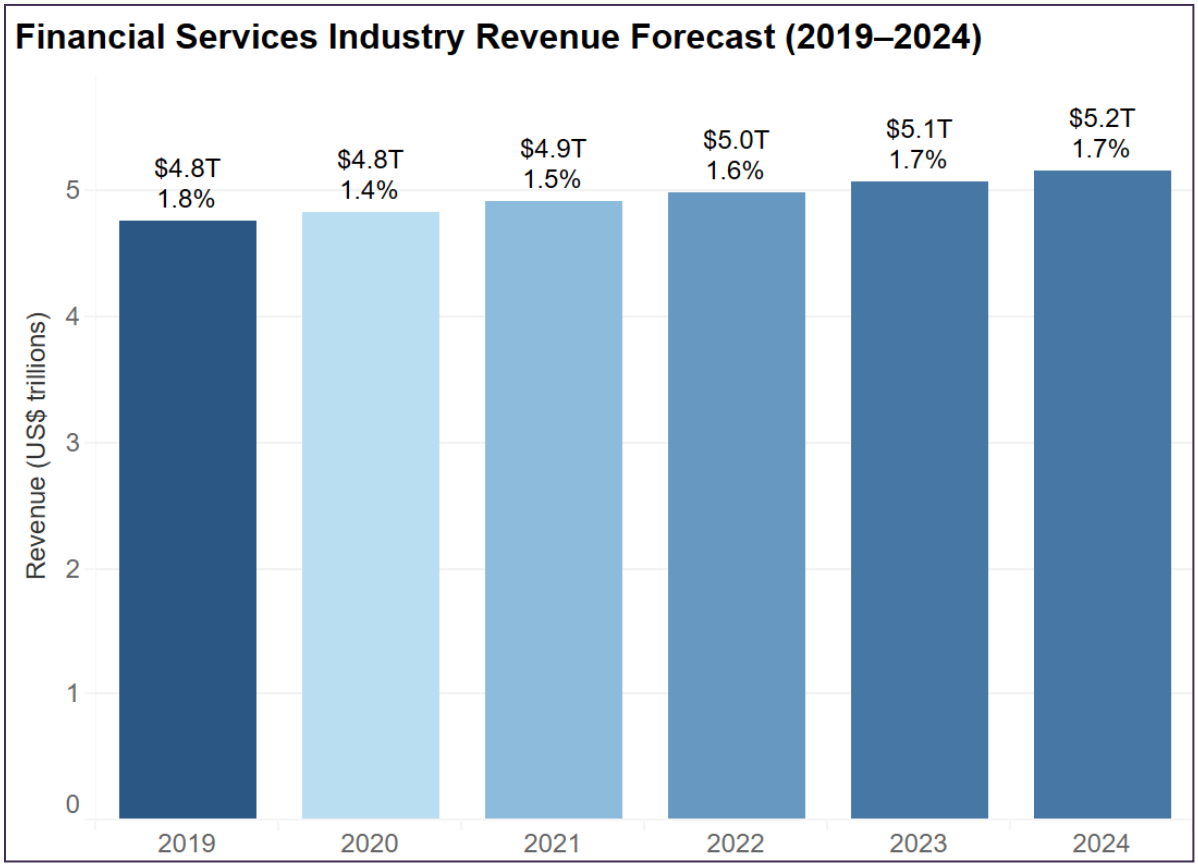

Figure 2. Financial Services Revenue Forecast from 2019 to 2024 [9]

Industry Segmentation

Within the financial services industry, there is a wide variety of operators providing numerous services to a variety of customers. The main services and products include the following activities:

- **Insurance and related activities.** Insurance is purchased by consumers and businesses to assure themselves of certain benefits in the event of a detrimental occurrence.
- **Credit intermediation and related activities.** Access to credit provided by operators in the sector is also critical for the economy to continue to function properly.
- **Securities, commodity contracts and other financial investments and related activities.** These operators, who provide securities, commodity contract, and other financial investments services, provide vital financing and savings services for consumers and businesses [9].

Figure 3 provides the approximate contribution of each sector to the overall industry revenue.

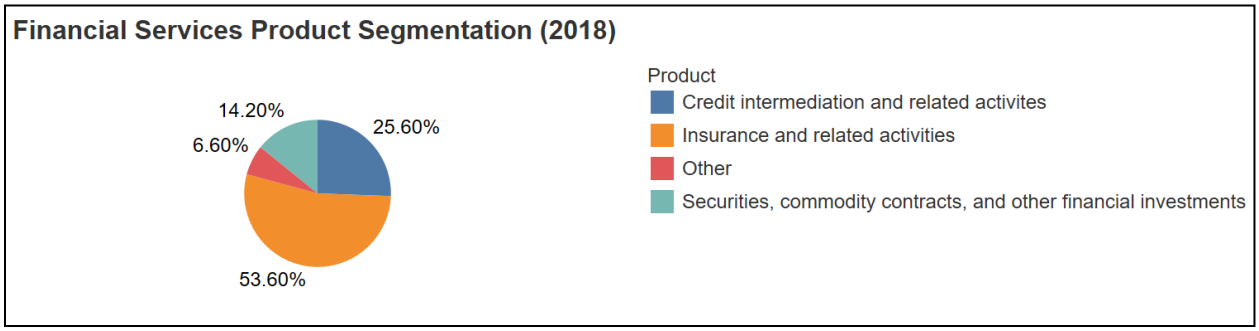

Figure 3. Financial Services Product Segmentation [9]

Insurance and related activities dominate the industry, accounting for 53.6% of the sector's overall revenue. Credit intermediation and related activities and securities, commodity contracts, and other financial investments make up 25.6% and 14.2%, respectively, of the industry's revenue. Other services make up just over 6% of the sector's revenue, accounting for the smallest portion of revenue [9].

Beyond the main activities listed here, the financial services industry includes transaction processing activities. Financial services entities are extensive users of electronic means for facilitating the verification of financial balances, authorizing transactions, transferring funds to and from accounts, notifying banks (or credit card issuers) of individual transactions, and providing daily summaries. Since these transaction processing activities are integral to the production of finance and insurance services, entities that principally provide a financial transaction processing service are classified to this sector, rather than to the data processing industry in the information sector [3].

Industry Geographic Segmentation

Operators within this sector tend to open operations within densely populated areas of the United States, usually in the immediate vicinity to potential clients [9]. See Figure 4 for business locations for entities within the financial services industry.

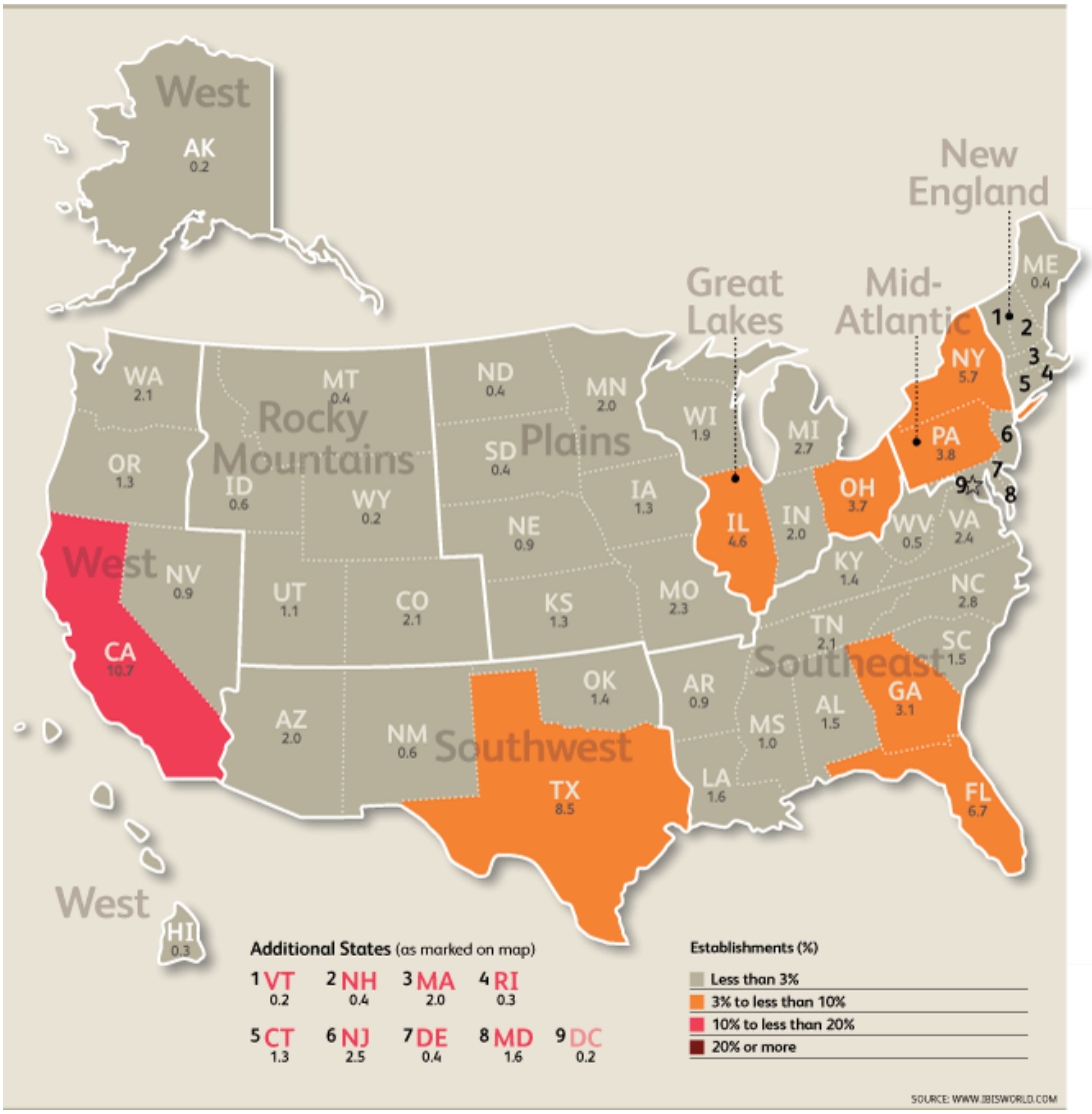

Figure 4. Financial Services Industry Business Locations in the United States. Image retrieved from IBISWorld.

About 25.4% of sector establishments are in the Southeast. The West is home to the second most sector establishments (15.4%), primarily located in California, which alone is home to 10.7% of sector establishments. Lastly, the Mid-Atlantic region is home to 14.2% of sector establishments [9].

Target Market Analysis: Online Payments Processing Services

Market Description

Online payments processing services comprise establishments (payments services providers (PSPs) or merchant services providers) primarily engaged in developing online payments processing software that enables merchants to authorize, settle, and manage credit card and electronic check transactions through the internet [8].

The primary activities of PSPs within this sector include developing online payment processing platforms, preventing and investigating fraud on online payment processing platforms, and developing mobile online payment processing platforms. The major products and services in this industry are mobile wallet payments, peer-to-peer (P2P) and money transfers, online bill payment, and online payment authentication [8]. Figure 5 below illustrates product segmentation within the market.

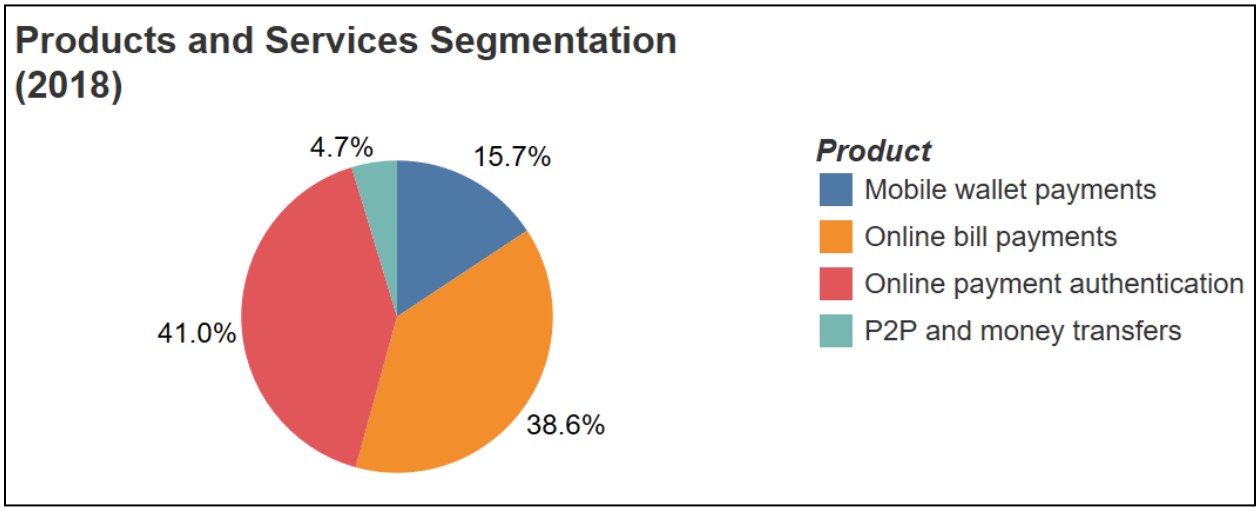

Figure 5. Online Payments Processing Services Product Segmentation [8]

Market Performance: Current and Historical

The online payments sector has experienced rapid growth over the period from 2014 to 2018 because of the expanding online marketplace and consumers' increasing level of comfort with purchasing goods and services online and through mobile applications. These trends coincided with growth in per capita disposable income, which has enabled consumers to purchase more products, necessitating the development of more payments-related services and products [8]. The payment processing market has also grown significantly because of the increased adoption of mobile wallets and instant access solutions that allow users to authenticate transactions using biometrics and to process payment transactions at a faster rate [7]. Figure 6 presents the revenue growth within the sector from 2008–2018.

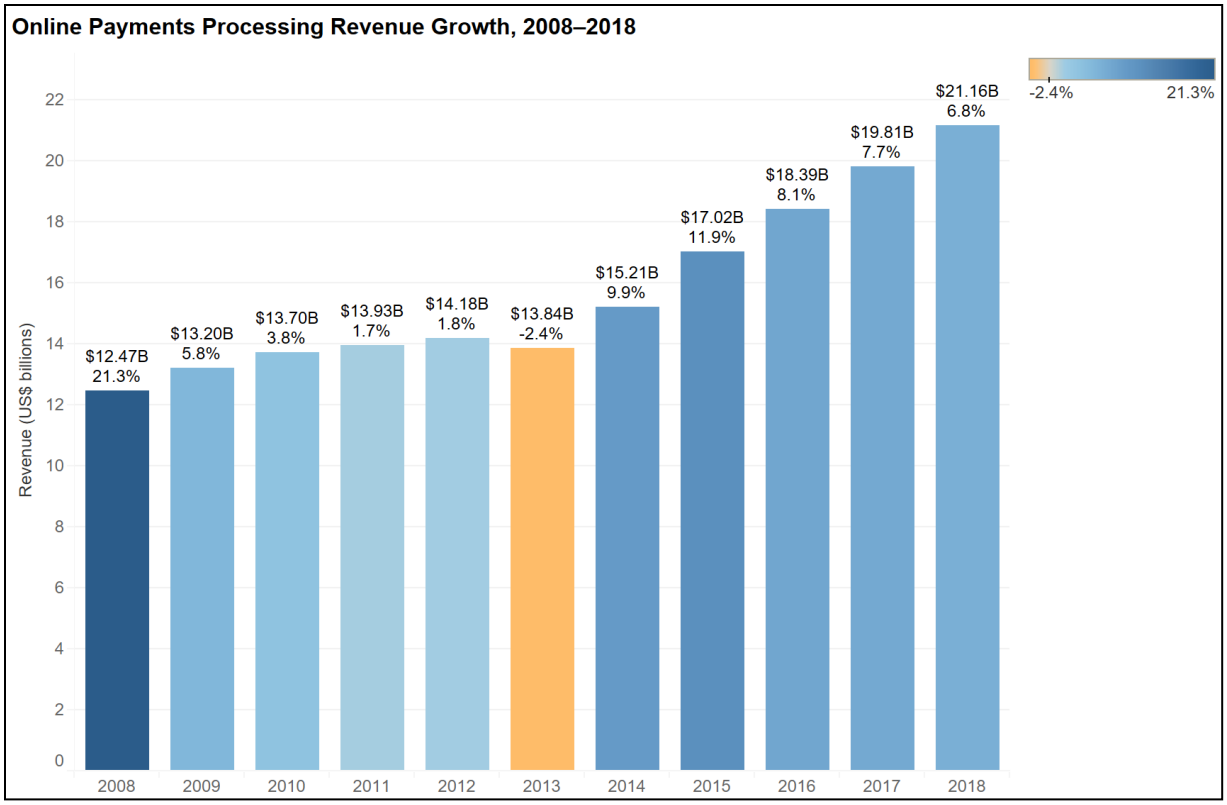

Figure 6. Online Payment Processing Revenue Growth from 2008 to 2018 [8]

Over the period from 2014 to 2018, revenue for the market has increased an annualized 8.9% to $21.2 billion in 2018, including growth of 6.8% in 2018 alone.

Market Outlook and Forecast

The payment processing solutions market is anticipated to grow at a rapid pace; the growing emphasis on digitization, increased use of online transactions, and expansion of the e-commerce industry are projected to propel the payment processing solutions market over the next six years [7]. Industry revenue is expected to grow an annualized 4.4% to $26.2 billion over the five years to 2023. Continued revenue growth will encourage new operators to enter the industry. As a result, the number of industry operators is expected to grow an annualized 4.0% to 1,008 over the five years to 2023 [8]. See Figure 7 for an illustration of the market's forecasted revenue from 2019 to 2024.

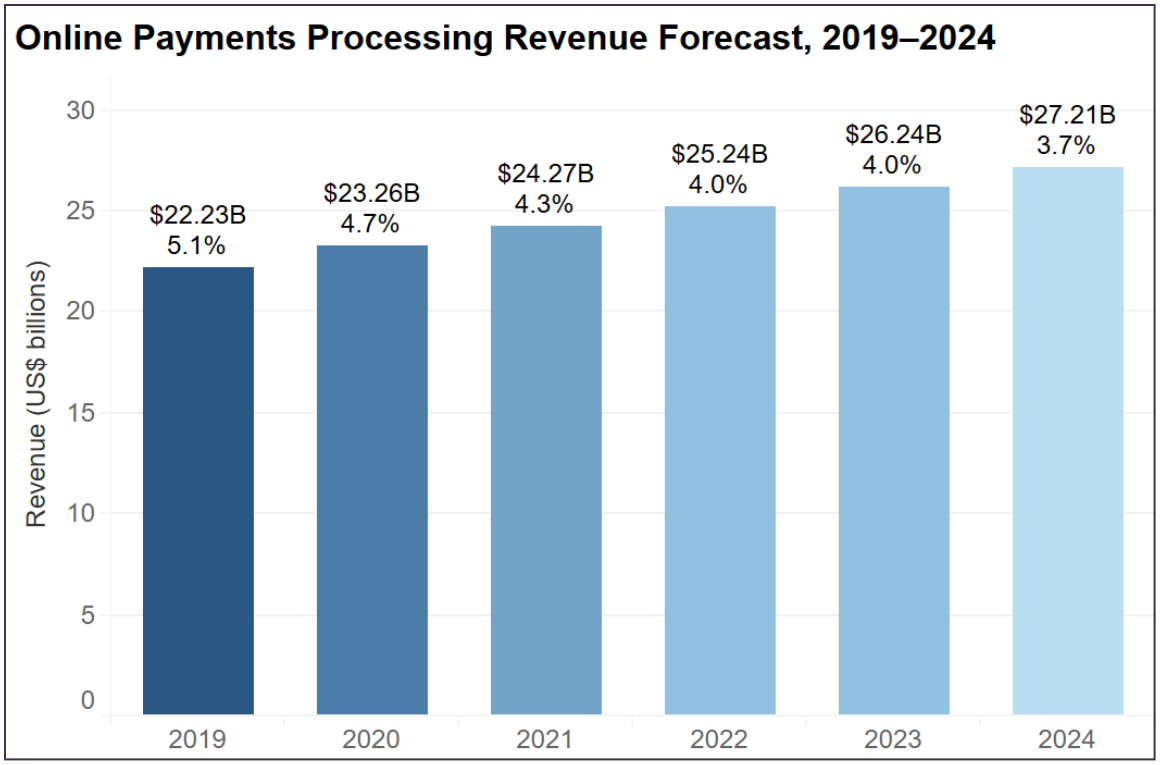

Figure 7. Online Payments Processing Revenue Forecast [8]

Market Segmentation

The online payments processing industry's primary market are merchants, which can be segmented into small, medium, and large merchants, that conduct business over the internet. Merchant business can be part of any industry that can function and sell services online. Additionally, a small portion of industry revenue comes from consumers that transfer money using payment processing software [8]. (See Figure 8.)

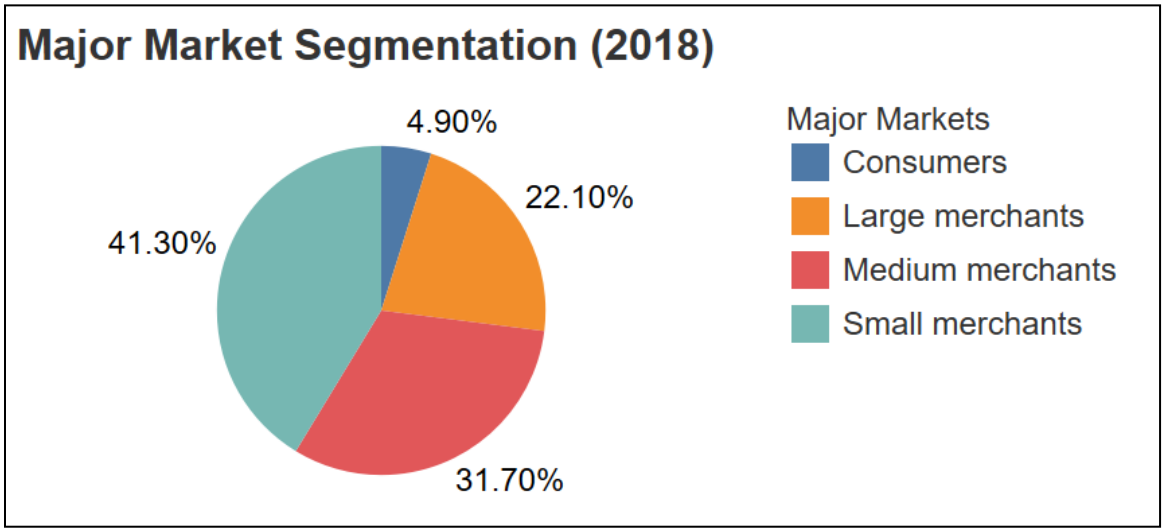

Figure 8. Major Market Segmentation for the Online Payments Industry (2018) [8]

Consumers account for a small portion of industry revenue at 4.9% in 2018. Consumers typically generate revenue for the industry when they transfer money to other consumers using their debit or credit card or when they transfer money outside of the United States [8].

Small merchants (businesses that generate less than $5.0 million in electronic sales volumes annually) are estimated to account for 41.3% of industry revenue in 2018. Pricing agreements with operators that take a percentage of each transaction are the most cost-effective way for these businesses to implement payment processing platforms on their sites [8].

Medium-sized merchants (business that generate between $5.0 million and $10.0 million in electronic sales volumes) account for 31.7% of industry revenue in 2018. Medium-sized merchants may find pricing agreements that involve an initial setup fee and recurring monthly fees more attractive than offerings of discounted percentage rates for higher sales volumes [8].

Large merchants (businesses generating more than $10.0 million in electronic sales volume) account for 22.1% of industry revenue in 2018. Large merchants benefit the most from discounted transaction fees as they process the most transactions of any sized merchant [8].

Market Characteristics

The online payments industry exhibits a high level of market share concentration and is driven by consumer adoption of online and mobile commerce and advances in digital payment technology.

Market Features

— **Large share of revenue controlled by established industry incumbents**. Such concentration in the market may be due to name recognition and consumers' trust in a well-known company's ability to keep their credit card information safe [8]. Some of the key players in this market are Stripe, PayPal Holdings, Inc., Amazon Payment, Inc., and WorldPay. Leading companies are focusing on strategic partnerships and mergers and acquisitions to gain competitive edge and increase market share [7]. Additionally, the industry's high level of concentration is indicative of a relatively new industry; the industry experienced rapid growth in the early part of the past decade, with many new entrants trying to capitalize on this growth. However, few have succeeded in capturing significant market share [8].

— **Increased competition.** Competition is increasing and is expected to force many entities out of the industry over the five years to 2023 [8]. New entrants must have highly skilled employees with fundamental knowledge of website development, online financial transactions, and cybersecurity. However, though a new payment platform may be secure, new entrants may struggle to gain the trust of merchants and consumers.

— **Complex payments ecosystem.** The processing of online monetary transactions

relies on a complex web of firms that includes payment acquirers/processors, independent sales organizations (ISOs), card networks, and issuing banks [1].

— **A high number of regulations.** As with all financial services providers, online payment processors are subject to laws regarding money transmission, fraud, and general banking:

  - Federal Bank Secrecy Act, which requires payment processors to prevent money laundering
  - Patriot Act, which requires payment processors to obtain licenses as money-transmitting businesses
  - Payment Card Industry Data Security Standard (PCI DSS), which requires all entities who process, store, or transmit credit card information to maintain a secure transaction environment or database
  - Digital Millennium Copyright Act (DMCA), which criminalizes the production and dissemination of technology, devices, or services intended to circumvent digital rights management
  - Various state money transfer laws [8]

— **Highly fragmented market.** Though large established operators control a large portion of industry revenue, the industry is highly fragmented, with most payment platform developers operating in specific verticals [8]. For example, certain payment processing developers may only target businesses of a certain size, while another company may develop software only for healthcare payments.

— **Somewhat volatile revenue streams.** Though the outlook for industry seems promising, revenue streams are affected by greater economic conditions. If consumers do not have the means to shop more, fewer transactions are processed, which can lead to a downturn in industry revenue.

Market Trends

The following trends have been observed in the online payments processing market:

— **Evolution of payments technology.** The concept of payments-as-a-platform (PaaP) is on the rise. PaaP connects disparate forms of payment platforms and other online payment methods, which would enable consumers to use any variety of payment options. Also, there is increased focus on the development of payments through near field communication (NFC), a short-range wireless connectivity standard that uses magnetic field induction to enable communication between devices brought within a few centimeters of each other [8].

— **Development of mobile payments and eWallets.** Much of the innovation and technological change (for example, developing NFC technology) occurring in the industry is centered on mobile payment platforms. Security issues tied to mobile commerce are also creating opportunities for new players to enter the industry to provide added security features to mobile payment processing platforms [8].

— **Disruption within peer-to-peer payments.** Online payments firms such as Venmo and Square are enabling payment functions once exclusive to cash and checks, access to which is controlled by banks [1].

— **Focus on security.** Theft has always been a central risk of the payments processing industry. Security of online and mobile payments have increased through advances in tokenization and biometric authentication. There also has been growth in the use of two-factor authentication to protect personal data in case a mobile device with payment information or an eWallet is lost or stolen [1].

Market Catalysts

The following are drivers of the payments processing market:

— **Faster and more reliable internet connectivity.** As internet connections have progressed and improved, many businesses have implemented e-commerce sites, which require some sort of payment processing platform. Additionally, an increase in the number of broadband internet connections in the United States and trends in household computer ownership influence demand for the industry's products [8].

— **Increased demand for mobile payment capabilities and the emergence of an expectation of omnicommerce** (the ability to pay with the same method whether buying in-store, online, or via a mobile device). As use of broadband-enabled mobile phones and tablets has grown, businesses are moving toward implementing mobile commerce platforms [8]. The shift toward omnicommerce requires retailers to adapt by allowing fast, simple, and secure mobile payments [5].

— **Enhanced technology and cybersecurity capabilities.** Technologies are providing new solutions that seek to increase the efficiency, accessibility, and security of financial services. New solutions, built on cloud and digital platforms, spanning mobile payments and peer-to-peer applications, have risen seeking to fill gaps [2].

— **Demand for data analytics on consumer behaviors.** Some large online businesses (e.g., eBay and Amazon) have implemented their own payments processing software. Through data analytics on the payments they are processing, and the consumers making those payments, these online companies hope to better understand of their consumers' behaviors and adapt their businesses toward those behaviors [8].

Competitive Landscape

Key Players

The major companies involved in the online payments processing market include PayPal, Stripe, Amazon Pay, and Square. Table 1 provides an overview of these companies:

Table 1. *Overview of Key Players in the Online Payments Processing Market and their Annual Revenue for 2018*

Company	About the Company*	Annual Revenue in USD Billions (2018)*
PayPal	PayPal's open digital payments platform allows users to connect and transact whether they are online, on a mobile device, in an app, or in person. Through a combination of technological innovation and strategic partnerships, PayPal creates ways to manage and move money and offers choice and flexibility when sending payments, paying, or getting paid. Available in more than 200 markets around the world, the PayPal platform, including Braintree, Venmo and Xoom, enables consumers and merchants to receive money in more than 100 currencies, withdraw funds in 56 currencies and hold balances in their PayPal accounts in 25 currencies.	15.8
Stripe	Stripe is a global technology company that combines economic infrastructure with a set of applications for new business models like crowdfunding and marketplaces, fraud prevention, and analytics. Businesses of every size—from new startups to public companies—use the company's software to accept online payments and run complex global operations. Today millions of companies across more than 100 countries use Stripe to start, run, and scale their businesses online. Stripe is currently focusing on helping more companies get started and thrive.	1.5
Amazon Pay	Amazon Pay makes it easy for millions of Amazon customers to pay using the information already stored in their Amazon account. It provides businesses with a familiar and convenient buying experience that can help their customers spend more time shopping and less time checking out.	265.5†

Square	Square's complete register service is a full point of sale with tools for every part of running a business, from accepting credit cards and tracking sales and inventory to small-business financing. Customers also use Square Cash, the easiest way to send and receive money, and Square Order, a new way to pre-order food and drinks for pickup. Founded in 2009 and headquartered in San Francisco, Square is currently available in the United States, Canada, and Japan.	3.9

* Source: Company profiles, Crunchbase. Available: https://www.crunchbase.com/
† Annual revenue data of Amazon Pay was not found. Annual revenue presented is for parent company Amazon.

Figure 9 to follow presents market share by number of U.S. websites using the leading payments platforms (including subproducts) in the market.

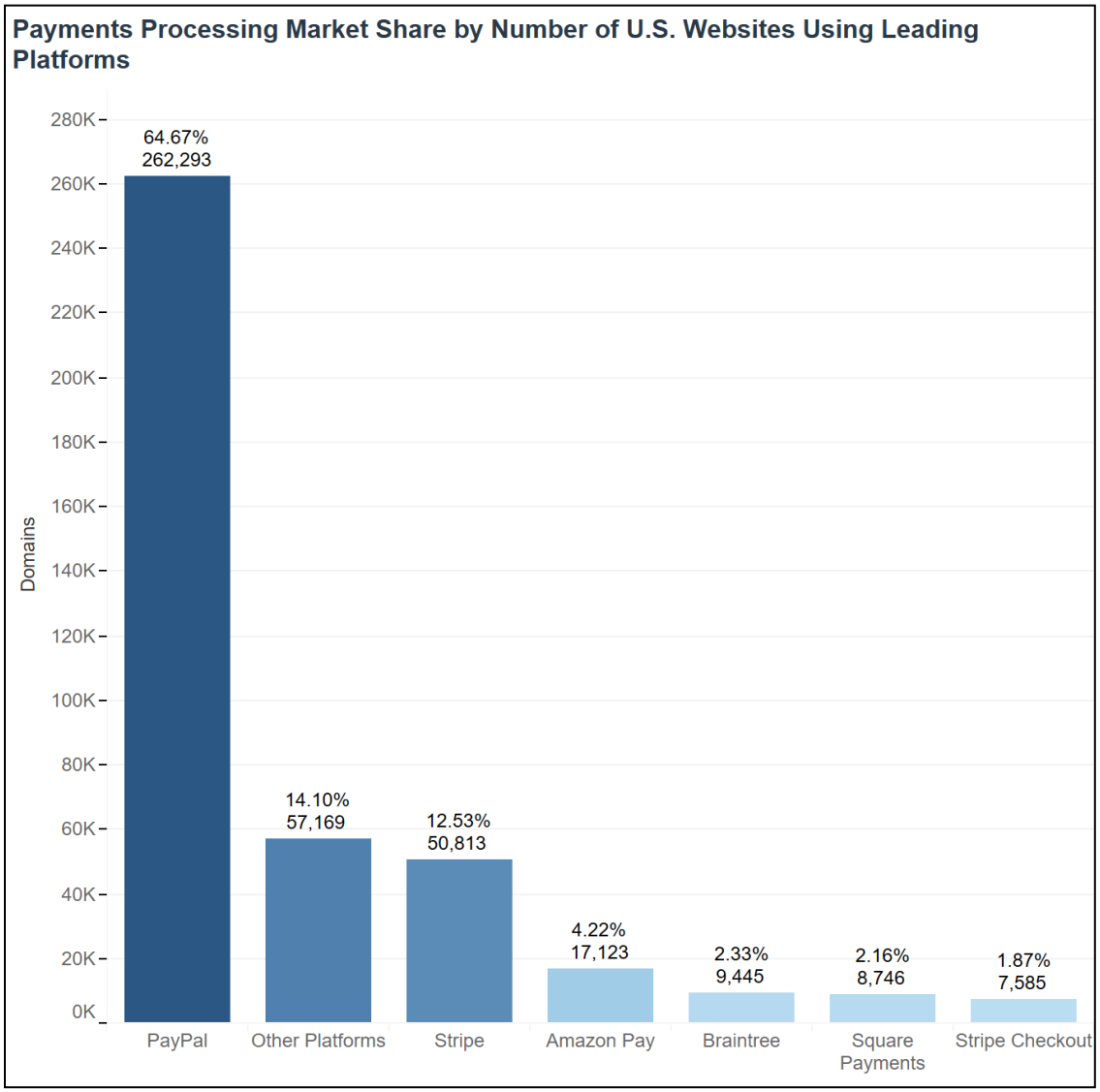

Figure 9. Payments Processing Platform Market Share by Number of U.S. Websites [6]

MerchBill's Competitive Advantage

Redevelopment of our recurring payments platform for subscription-based online merchants can provide us a competitive advantage in the online payments processing market in the following ways:

— **Be at the forefront of payments technology.** Implementation of the latest technology available can improve transaction efficiency and lower operational costs [8]. An update to our payments technology will allow us to offer detailed analytics to track ongoing subscriber trends and integrate add-ons such as customer acquisition solutions and campaign management facilities.

— **Boost brand awareness.** A reconfiguration of our product will help us increase market share with merchants we have yet to reach, such as streaming services and fashion subscription boxes, and we can build our brand name through our relationships with subscription-based business. Platforms that have a strong foundational merchant base typically experience stronger adoption rates of their service [8].

— **Develop consumer trust.** A flexible and frictionless payments experience through the new system will help us obtain merchant partners since it will enhance the consumer experience and help the merchants retain customers.

Limitations to the Status Quo

Limitations contributing to a status quo in online billing platforms indicate that priority should be placed on investing in and developing payments technology for the future.

Overview of Online Billing Platforms

An integral part of e-commerce and online-based businesses, an online billing platform or system refers to software that allows merchants to collect customers' names, billing details, and terms of payment as well as product or service information. When processing payments, the system must ensure a high level of security and compliance with all credit card and banking regulations. Subscription billing (recurring billing) adds another layer of complexity to processing online payments, requiring the system to ensure that recurring payments are processed on the correct date for the agreed-to amount through the appropriate payment method. Billing solutions can either be built in-house or are purchased (typically, through a Software-as-a-Service [SaaS] model) and implemented into merchants' systems through application programming interfaces (APIs).

Limitations to Online Billing Platforms

In the current payments industry, merchants often choose larger, well-known payments services providers such as PayPal; however, the market is highly fragmented with many payment platform developers functioning only in certain verticals. This environment leads to the following issues:

— **Lack of subscription management capabilities**. Typical online billing platforms offer the ability to process recurring payments, but don't offer features important to subscription-based merchants such as tracking the subscription lifecycle of an account or measuring operational metrics such as recurring revenue and churn statistics. However, companies that specialize in services to subscription businesses may offer such account management features.

— **Difficult to implement and customize.** Integration and customization of an online billing solution may require more expertise than a merchant has access to in house. Often, billing platform providers publish detailed API guidance for merchants' developers, but merchants may require further support services. Some billing platform providers offer implementation services for an added cost.

— **Lack of flexibility.** Some billing solutions fit the needs of small companies, while others are built to handle the complexities of large companies. Rarely is there a billing solution that grows with a company, from startup stage to major enterprises.

— **Hard to switch billing solution providers.** Once a merchant uses a certain billing platform, it may find it difficult or too expensive to switch to a new solution though it may be a better fit. A merchant may feel that too much money and effort was spent on implementing the old billing platform to end the relationship with their current billing platform provider.

— **Regulatory complexity.** The billing system must adhere to the technical and operational requirements of the Payment Card Industry Data Security Standard (PCI DSS). The purpose of the standard is to promote and improve cardholder data security and facilitate the adoption of a standard for global data security measures. PCI DSS applies to all entities involved in payment card processing and to entities that store, process, or transmit cardholder data or sensitive authentication data [15].

— **System or payment gateway failure.** A payment gateway is the system that securely sends payment information from a merchant to the credit card network or bank and returns transaction details and responses from the card network or bank. If a merchant's specified payment gateway or billing system suffers an outage or other problem, transactions cannot be processed, leading to lost revenue.

Significance of Limitations on Stakeholders

Merchants need reliable, PCI-compliant online billing systems that offer data analytics and customer management capabilities to run their businesses efficiently; customers rely on these systems to securely process their payments for goods and services. However, the limitations of traditional billing systems impact these stakeholders:

Impact on merchants. Lacking the expertise to build a billing solution in house or customize a platform with unwieldy APIs, small- and medium-sized merchants often must choose an online billing system that does not suit their needs or does not offer the capabilities that will help them grow. Subscription business models offer a wealth of data on customers, but if the billing software doesn't track that data, merchants are losing vital business insight.

Impact on consumers. Security and privacy are the most important concerns for consumers when it comes to processing payments. If the payments platform is not secure or PCI compliant, consumers' private financial information might be leaked.

Significance for MerchBill

Currently, our platform suffers from some of the aforementioned limitations. Our platform does offer flexibility, but merchants need to have a high level of developer expertise in order to customize the platform to their systems, which leads to weak merchant adoption by smaller businesses. We also don't offer automated subscription management capabilities that can help merchants grow. Tackling these limitations all at once would be difficult, time consuming, and expensive. With our strong foundation of payments capabilities, we can quickly rebuild part of our payments platform to offer automated subscription management capabilities while keeping an eye toward future updates.

Opportunity Fulfillment: Push Notification Microservice and Dunning Management

System Architecture

Service Overview

MerchBill's SMB platform supports a broad range of capabilities that merchants need for subscription billing, including self-deploy or turnkey implementation, billing administration, integrated fraud monitoring, and a merchant portal with customized access roles. Capability enhancements to the current product include a cloud-based push notification microservice and integrated dunning management. The SMB platform is available in three pricing plan options, each tailored to fit a revenue growth stage. Merchants can choose the subscription lifecycle management plan that best suits their business size and needs.

The following sections discuss the functions of the push notification microservice and dunning management features as well as a detailed list of the platform's features and functionality.

Key Feature: Push Notification Microservice

Recurring payment activity, such as invoice generation, attempted payments, and subscription cancellations, are handled by our SMB platform. The push notification microservice would notify a merchant when these events take place so that they may take any account management steps. The most common items that merchants would need to be notified of are:

— **Payment failures.** A payment failure may be because of insufficient funds, outdated payment details, or a card issuer refusing the charge. A merchant can rely on push notifications to inform them of payment failures on subscription accounts. Notification of payment failure can also support dunning management, which is discussed further in the next section.

— **Payments that require additional action.** Sometimes additional information from or action by the customer is required to obtain payment. When such circumstance occurs, a merchant can receive notification of the event so that they can contact the customer.

— **Subscription lifecycle changes.** Through push notifications, merchants can track subscription status changes, such as trial, active, paused, or canceled.

Merchants will have the ability to configure push notifications and subscribe to only those that are meaningful to them. Account event notifications they can subscribe to include:

— **Prebilling** – Sent within 30 days (based on merchant preference) of when subscription payment is to be processed

— **Failed payment notice** – Sent when payment cannot be processed either because of a declined payment or an error response from the processor

— **Payment failed on retry** – Sent when payment fails on a subsequent payment attempt

— **Credit card about to expire** – Sent within 30 days of card expiration based on merchant preference, for example, notification can be sent 5 days before card is about to expire

— **End of rights entitlement** – Sent when customer rights or access to service or product ends

— **Account status change** – Sent whenever there is a change in account status such as trial, active, paused, or canceled

These push notifications also facilitate integrated dunning management, which is discussed in the next section.

Key Feature: Integrated Dunning Management

Dunning refers to requesting payments of a debt. Dunning management is the ability to handle the process of making those requests. An integrated dunning management system would allow merchants to:

— **Efficiently respond to declined payments.** A merchant can customize automatic actions to take in response to being notified about a declined payment.

— **Retry payment methods on file automatically.** A merchant can choose when to retry a credit card after a failed payment rather than suspending the account the first time the payment is declined.

— **Send automatic email communications to customers.** Merchants can automate customer communications regarding failed payment through email templates and tailored schedules.

— **Track number of successful and failed transactions.** A merchant will know the number of failed payments versus successful payments, which is data that can be used to inform the merchant’s overall business strategy.

Features and Functionality

The following table lists the features and functionality of the SMB platform:

Table 2. *Summary of Main SMB Platform Features and Functionality*

Feature	Description
Subscription Management	— Create, manage, pause, cancel, reactivate, and extend subscriptions — Handle upgrades and downgrades — Keep customers engaged with automated emails (including dunning communications) — Set rules around customer notifications — Customize trial periods, extend trials indefinitely, and offer free/paid trials — Customize renewal dates for customers based on requests
Billing Administration	— Easily access and change key billing functions through a self-service portal — Add, update, and delete accounts, billing plans, and products — Manage pricing plans — Support payment failure analysis, management, and notification — Update customer billing information — Change payment methods or billing frequency — Issue refunds — Cancel accounts because of chargebacks or customer requests — Retry payment methods after initial payment failure
Global Transaction Support, Scalability, and Reliability	— Hardware redundancy for all components — Physical and network redundant infrastructure across geographic regions — Support for multiple payment methods, including credit cards, PIN-less debit cards, automated clearinghouse, and stored value cards — Support for multiple payment processors and worldwide currencies — Ability to send out customized billing messaging in multiple languages — Native tax engine with support for merchant connection and appropriate goods and services tax, value-added tax, or sales tax

Security and Compliance	— Cryptographically secured permissions, roles, and responsibilities — Strong encryption protocols certified to FIPS 140-2 — Real-time fraud screening to determine chargeback probability — Built-in chargeback management to recover lost revenues and determine root causes of fraud
Merchant Notifications	— Set account notifications based on merchant preference — Subscribe to notifications for the following events: - Prebilling - Failed payment notice - Payment failed on retry - Credit card about to expire - End of rights entitlement - Account status change
Reporting	— Pull reports focused on financial and business metrics — Track affiliate revenue and payments — Track successful and failed payments — Download reports for integration with marketing analytics

Benefits of the New SMB Platform Features

Benefits for Merchants

The following table presents merchant benefits from dunning management and the notification microservice:

***Table 3**. Summary of Merchant Benefits*

Benefit	Description
Frictionless Payments and Revenue Recovery	— Lower churn and increase revenue with automatic charge retries — Support overall business strategy though tracking of account statuses and payment processing success rates
Automated Account Management	— Simplifies merchants' management of customer accounts — Provides a mechanism for merchants to synchronize data within payments platform, customer relationship management platform, and internal systems — Allows for automatic responses to customers regarding payment status — Allows merchant to intervene before credit card expiry

Customer Relationship Management	— Allows merchants to focus on building relationships, not just transactions since they can spend less time on addressing declined payments or billing-related functions — Allows merchants to address active customer turnover with discounts and loyalty offers through proactive communications

Benefits for MerchBill

The following table presents benefits for MerchBill from the new SMB platform features:

Table 4. *Summary of Benefits for* MerchBill

Benefit	Description
Viability in the Market	— Builds out brand name through new offerings — Diversifies our client base and increases market share with merchants we have yet to reach since we can offer enhanced account management features
Reduced Internal Costs	— Reduces maintenance burden to engineering — Allows for fewer database calls that put stress on the system — Supports general application stability
Technological Agility	— Supports extensibility of the platform — Allows for faster time to market for future notification features

Implementation Plan

Feasibility Assessment

Technical Feasibility

Though the new features will require a complete refactor of a portion of the SMB platform, we will be building with the future in mind by using a cloud-based microservice leveraging Amazon Web Services (AWS) and rewriting existing code from Perl to Python. MerchBill already has the staff knowledgeable in Python and cloud architecture to implement the proposed features. Also, little training from the senior software engineer to the rest of the development team should be required.

Financial Feasibility

Since MerchBill already has the required professionals and technical knowledge for this project, the SMB platform enhancement costs will be minimal. The estimated expense of the project over 10 months is about $1.4 million. The value of the new features

should outweigh implementation costs and will potentially increase our revenue from the SMB platform.

Marketability

For MerchBill, the merchant adoption of subscription-based business models presents an opportunity to offer our billing services to more merchants and presents the risk of falling behind our competitors. Managing the subscription billing lifecycle can be complex, and merchants look for the following in subscription billing platforms:

— **Ability to recover revenue and reduce customer churn.** Merchants need to be able to handle declined transactions. When merchants can salvage failed transactions, they can reduce the number of customers they lose.

— **Secure and compliant payments system.** Merchants' customers require that their information is kept safe and that storage of all data is held to applicable industry and government standards.

— **Customer data analytics.** Subscriptions offer merchants the ability to monitor customer behaviors overtime. A good subscription billing and management platform should allow merchants to see important insights from the stored subscription data.

— **Flexibility with reduced complexity through technology.** The subscription billing platform should be able to grow with the merchant's business and offer ease of use through a comprehensible API. The system should also handle typical payments events such as automatic merchant notifications and email alerts to customers.

The proposed platform enhancements will allow us to respond to merchant requirements for reduced customer churn and customer management support.

Recommendation

MerchBill must move with the industry and offer our subscription billing and management services to new entrants of the subscription economy and to those merchants that are growing. In order to have a product that merchants respond to, we must add new features to refresh our platform. However, we have the added challenge of overcoming our competitors.

Our Challenge and Competitors

MerchBill is a relatively smaller player in the overall payments processing industry with an estimated annual revenue of about $18.3 million [4]. In comparison, the leader in the payments processing industry, PayPal, pulled in an annual revenue of about $15.8 billion. As a small start-up, MerchBill primarily competes with other small subscription billing and management companies such as Zuora, Aria Systems, and Recurly. See Figure 10 for an illustration of MerchBill's revenue compared to our competitors.

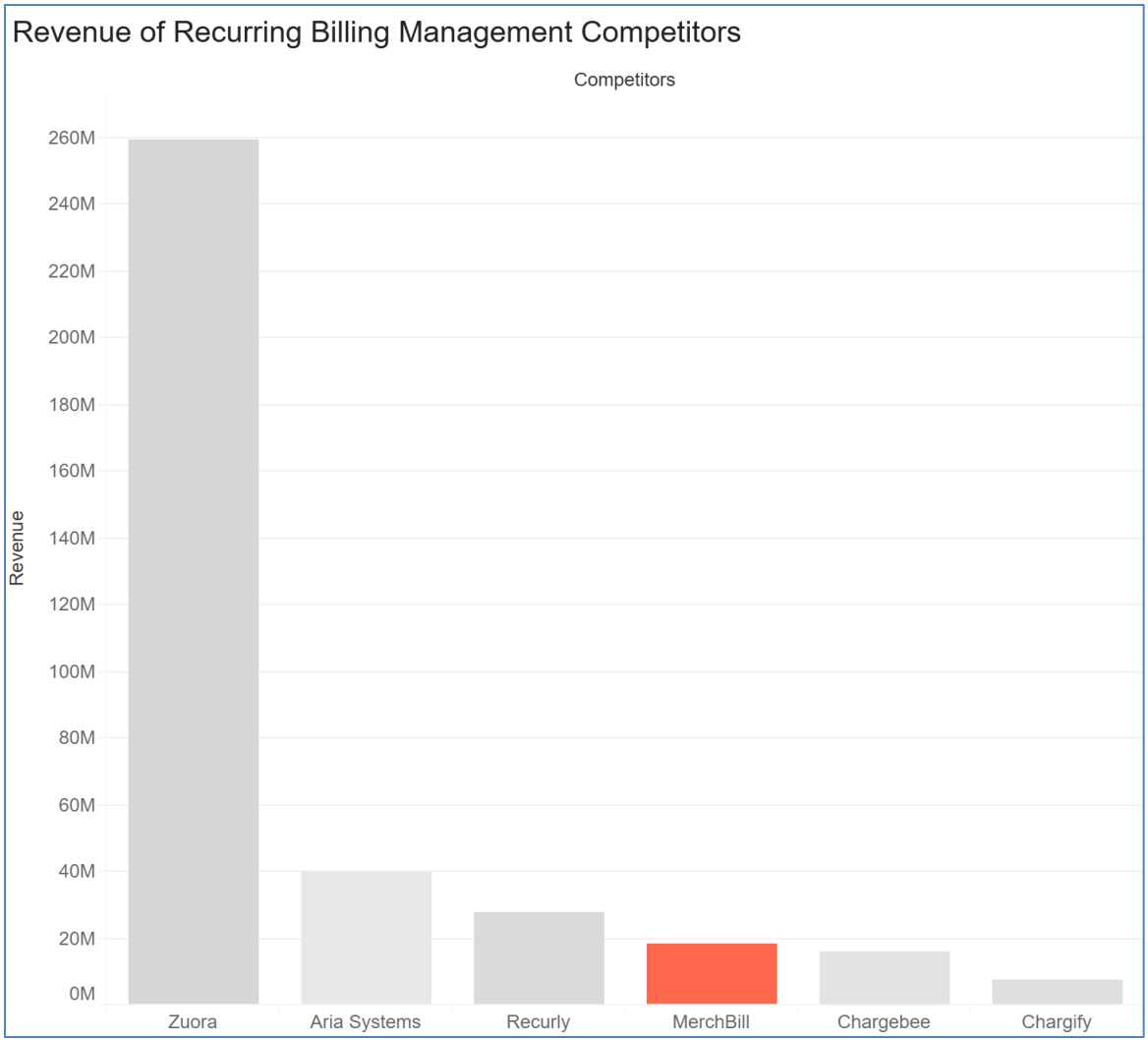

Figure 10. Annual Revenue of MerchBill Compared to Competitors [4]

A Refreshed Platform

With the blossoming of subscription-based business, we can show new merchants and old that we are able to meet their subscription needs and become a leading provider within the subscription payments industry. According to Forrester Wave, leading recurring billing management platforms improve business agility for their merchants to react to changing market conditions; automate consequences to subscription lifecycle changes; and provide actionable, recurring customer- and revenue-focused insights [12]. Our current platform provides the flexibility that merchants desire, and merchants can also launch campaigns and test pricing through marketing and sales components. However, there are a few areas in which our current SMB product is lacking:

— **No notifications are sent to merchants regarding subscription status.** Our current platform requires the merchant to constantly call on our database to check account status, and they don't receive notification when an account's status changes due to payment failure.

— **No integrated dunning management.** There is no automatic communication with customers about declined payments; merchants must spend time sending email reminders to customers whose credit cards have been declined.

— **Integration is difficult.** Our APIs don't allow developers to build easily on top of the platform, which in turn might not be the best option for companies that want to do their own customization.

To progress forward, we must quickly adapt to meet merchant needs and to improve transaction efficiency and lower operational costs. To begin, we can improve our platform by adding two key features: push notification microservice and dunning management.

Resources Required

MerchBill currently employs all required staff to begin implementation. Only the following resources will be required:

Personnel

Required personnel for implementation includes:

— One product manager to distribute tasks and manage development team

— Three software development engineers to cultivate new features, fix bugs, provide project status, and do research

— One quality assurance tester to verify proper component functionality and integration

— One technical writer to produce API documentation.

Financial and Physical Considerations

MerchBill's cost of implementation and feature development for this project will not exceed $1.4 million. See Table 4 for more details. The primary expense will be toward the development team salaries, which is an annual expense of $690,000. A marketing campaign highlighting our new features will cost about $400,000, and miscellaneous costs will be approximately $300,000.

Time Frame

All implementation work will be completed within 10 months.

Implementation Timeline

Implementation of the push notification microservice into the main SMB platform will be broken up into three phases, spanning a total of 10 months. Figure 11 illustrates the implementation's target timeline.

Implementation Phases	Month 1	Month 2	Month 3	Month 4	Month 5	Month 6	Month 7	Month 8	Month 9	Month 10
Phase 1: Message Service	3 months									
Phase 2: Webhook Management				3 months						
Phase 3: Event Service							2 months			
Phase 4: Dunning Enhancements									2 months	

Figure 11. *Implementation Timeline for SMB Platform Enhancements*

Financials

MerchBill's annual expenses required for the SMB platform enhancements are detailed in the table below.

Table 4. *Detailed Outline of MerchBill's Required Financial Expenses for the SMB Platform Enhancements*

Financial Application	Annual Unit Cost (USD)	Quantity	Annual Cost (USD)
Personnel			
Product Manager	150,000	1	150,000
Senior Software Engineer	130,000	1	130,000
Software Engineer	100,000	2	200,000
Quality Assurance Analyst	110,000	1	110,000
Technical Writer	100,000	1	100,000
Subtotal			**690,000**
Marketing			
Advertising	400,000	1	400,000
Subtotal			**400,000**
Miscellaneous			
Office space	240,000	1	240,000
Office equipment	60,000	1	60,000
Subtotal			**300,000**
Total			**1,390,000**

Development Team

Development Team Overview

The development team for the SMB platform enhancements is made up of skilled individuals who want to characterize MerchBill as a leading provider of subscription management and billing services.

The development team includes:

— **Sean Clawman:** Product Manager

— **Chris Meowfee:** Lead Software Engineer
(Two software engineers to be determined.)

— **Rachel Purrenga:** Quality Assurance Senior Manager

— **Priscilla Miller:** Technical Writer

Development Team Profiles

Sean Clawman

Product Manager

Knowledgeable in PCI compliance and AWS technologies, Sean has 4 years of experience advancing new ideas and development efforts for products closely aligned with customer needs and requirements. Prior to his role as a product manager, Sean worked a software engineer for SmartyCats. Sean obtained a bachelor's degree in computer science from Notre Dame de Namur University and is Scrum Product Owner certified.

Chris Meowfee

Senior Engineer and Cloud Architect

Specializing in Rest APIs, cloud systems, and security, Chris is a lead software engineer with 3 years of experience. As a senior engineer, he is responsible for developing and maintaining software code in accordance with technical design specifications. Chris has a bachelor's degree in computer science from the California Institute of Technology.

Rachel Purrenga

Quality Assurance Analyst

Rachel has nearly 10 years of experience in partnering with both business and technical teams to assist during product development testing activities and creating and executing test plans and test cases to match business requirements and thoroughly test software to ensure proper operation. Prior to joining MerchBill nearly 5 years ago, she worked as a quality management specialist at Catnip Inc. Rachel earned her bachelor's degree in computer science at the University of Washington and has also earned a Certified Quality Auditor certification from the American Society for Quality.

Priscilla Miller

Technical Writer

Experienced in collaborating with subject matter experts to support documentation development, Priscilla is a technical writer with 1 year of experience. She is skilled in documenting operating procedures, user instructions, and other materials for all phases of product development. Priscilla obtained a bachelor's degree in English literature from Seattle University and is also working toward a second bachelor's degree in information technology from the University of Massachusetts – Lowell. She is also participating in a professional sequence in technical communication through UC Berkeley Extension.

Appendix: Glossary of Terms and Abbreviations

API	Application programming interface
AWS	Amazon Web Services
chargeback	A return of money to the consumer
churn	Customer attrition; the loss of clients or customers
dunning	The process of communicating with customers to ensure the collection of debts owed
e-commerce	The buying and selling of products and services over the internet; short for electronic commerce
fintech	The integration of technology into offerings by financial services companies in order to improve their use and delivery to consumers
FIPS 140-2	Federal Information Processing Standard Publication 140-2, a U.S. government computer security standard used to approve cryptographic modules
fraud	Wrongful or criminal deception intended to result in financial or personal gain
invoice	A document issued by a merchant to a consumer relaying the agreed prices for products or services provided by the merchant to the consumer

merchants	Small, medium, and large business that conduct business over the internet selling goods and/or services
microservice	A service that is part of a larger microservice architecture, which structures an application as a collection of services
payment gateway	An application service that authorizes credit card or direct payments processing for merchants
PCI DSS	Payment Card Industry Data Security Standard
platform	A major piece of software under which various smaller application programs can be designed to run
PSP	Payments services providers
push notification	An automated message sent by an application to a user when the application is not open
recurring billing	Automatic charging of a cardholder for specified goods or services on a prearranged schedule
recurring revenue	Revenue that is expected to be received on a regular basis
rights entitlement	Authorization to use, access, or consume an application or service or product
SaaS	Software-as-a-Service

SMB	Subcription management and billing
subscription	Agreement for a customer to pay a merchant for products or services for a specified time-period
subscription lifecycle	The various stages of a subscription, from creation to cancellation
subscription management	Coordination of customer lifecycle operations such as managing trials, assigning credits, issuing refunds, and making account changes
trial	Testing of a product or service for a consumer to assess if they should continue using the product or service
webhooks	A way for an app to provide other applications with real-time information

Reference List

[1] "An introduction to fintech: Key sectors and trends" S&P Global Market Intelligence (October 2016). Available: https://www.spglobal.com/marketintelligence/en/documents/an-introduction-to-fintech-key-sectors-and-trends.pdf>

[2] "Fintech: The Experience So Far" International Monetary Fund (17 May 2019). Available: https://www.elibrary.imf.org/doc/IMF007/26184-9781498321860/26184-9781498321860/Other_formats/Source_PDF/26184-9781498322744.pdf?redirect=true>

[3] "2012 NAICS: 52 - Finance and insurance" Industry Statistics Portal, U.S. Census Bureau. Available: https://www.census.gov/econ/isp/sampler.php?naicscode=52&naicslevel=2#

[4] Company profiles, Crunchbase. Available: https://www.crunchbase.com/

[5] "Overview of the Payments Industry" Medici (31 January 2018). Available: https://gomedici.com/overview-of-the-payments-industry

[6] "Payment Processing Profile" Datanyze. Available: https://www.datanyze.com/

[7] "Payment Processing Solutions Market Size, 2019 To 2025, Report Summary" Grand View Research. Available: https://www.grandviewresearch.com/industry-analysis/payment-processing-solutions-market

[8] Gambardella, Anthony, "U.S. Specialized Industry Reports – Online Payment Processing Software Developers" IBISWorld (December 2018). Accessed: Nov. 14, 2019. [Online].

[9] Gambardella, Anthony, "U.S. Industry Reports (NAICS) – Finance and Insurance" IBISWorld. (May 2018). Accessed: Nov. 14, 2019. [Online].

[10] "Finance and Insurance: NAICS 52" Bureau of Labor Statistics. Available: https://www.bls.gov/iag/tgs/iag52.htm

[11] Schmidt, Chris. "Recurring Revenue Rising" CFO (27 February 2019). https://www.cfo.com/cash-flow/2019/02/recurring-revenue-rising/

[12] Varon, Lily. "Recurring Customer And Billing Management, Q3 2017" The Forrester Wave (31 July 2017).

[13] Chen, Tony; Fenyo, Ken; Yang, Sylvia; and Zhang, Jessica. "Thinking inside the subscription box: New research on e-commerce consumers" McKinsey & Company (February 2018).

[14] Longanecker, Chuck. “Why You Should Use a Subscription Business Model” *Entrepreneur* (March 19, 2015). Available: https://www.entrepreneur.com/article/243573

[15] Turner, Dawn. "PCI DSS requirements for building and maintaining a secure network and systems" Utimaco (October 19, 2019). Available: https://content.hsm.utimaco.com/blog/pci-dss-requirements-for-building-and-maintaining-a-secure-network-and-systems

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